

Technical Design

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Project Name: AI-Powered Knowledge Assistant

Team Name: Hello World

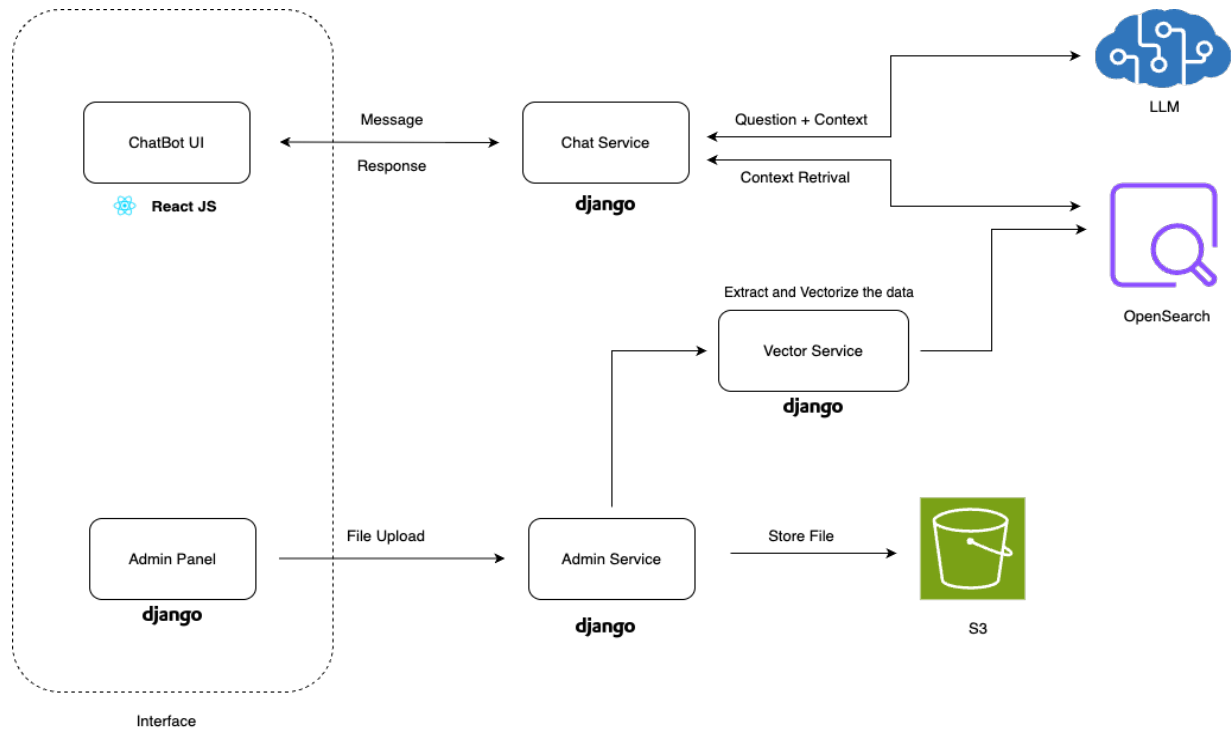
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Document Status:

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AI Powered Knowledge Assistant

System Components

- **Frontend Components:**
 - React-based Chat Bot (embedded as an iframe)
 - Admin Panel in Django
- **Backend Services:**
 - Django REST APIs (Chat & Admin Service)
 - Upload, parse, and vectorize modules
 - LLM answer generation and OpenSearch query module
- **Database Design:**
 - AWS OpenSearch Service for vectorized chunks
 - Files stored in AWS S3
- **Integration Points:**
 - AWS S3 (File storage storage)
 - AWS OpenSearch service (vector DB)
 - Hugging Face Transformers (nomic embedding)
 - Qwen3 Open source LLM

Technology Stack

- **Programming Languages:** Python, JavaScript
- **Frameworks & Libraries:** Django, React JS, NLTK, PyTorch, FAISS
- **Database Technologies:** OpenSearch , S3
- **Cloud Services:** AWS S3, OpenSearch Service, EC2
- **Development Tools:** Git, GitLab, Postman, Docker, VS Code, PyCharm

Data Flow & Processing

- **Data Input Sources:** User-uploaded PDFs or .txt files
- **Processing Logic:**
 - Extract text
 - Sentence tokenize
 - Chunk + embed with Nomic model
- **Data Storage Strategy:**
 - File in S3
 - Chunk vectors in OpenSearch
 - Metadata in OpenSearch fields
- **Data Output/APIs:**
 - /upload/ for upload
 - /message/ for Q&A

Security Design

- **Authentication Methods:** JWT-based user sessions (planned)
- **Authorization Framework:** Role-based ACL (future)
- **Data Encryption:**
 - HTTPS endpoints
 - S3 uses SSE-S3 or KMS for encryption
- **Security Protocols:**
 - IAM for S3 access
 - OpenSearch Basic Auth + HTTPS

Performance Considerations

- **Response Time Targets:** < 5 seconds per query
- **Scalability Strategy:** Stateless services, async embedding jobs
- **Optimization Approaches:**
 - Pre-caching frequent vectors
 - Embedding reuse from past queries